

PROOF OF LEMMA 5

The vector $\bar{\mathbf{w}}^{t+1}$ can be rewritten as in the following.

$$\begin{aligned}\bar{\mathbf{w}}^{t+1} &= \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} \mathbf{w}^{L,m,u,t}, \\ &= \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} (\mathbf{w}^{L-1,m,u,t} - \eta_t \tilde{\nabla} f_{m,u}(\mathbf{w}^{L-1,m,u,t})),\end{aligned}\tag{38a}$$

$$= \mathbf{w}^t - \eta_t \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} \sum_{l=0}^{L-1} \tilde{\nabla} f_{m,u}(\mathbf{w}^{l,m,u,t}) \right],\tag{38b}$$

$$\begin{aligned}&= \mathbf{w}^t - \eta_t \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} \sum_{l=0}^{L-1} (\tilde{\nabla} f_{m,u}(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{b}^{l,t}) + \nabla f(\mathbf{b}^{l,t})) \right], \\ &= \mathbf{w}^t - \eta_t [\mathbf{e}^{0,t} + \nabla f(\mathbf{b}^{0,t}) + \dots + \mathbf{e}^{L-1,t} + \nabla f(\mathbf{b}^{L-1,t})], \\ &= \mathbf{w}^t - \eta_t (\mathbf{g}^t + \mathbf{e}^t).\end{aligned}\tag{38c}$$

where (38a) comes from Step 8 of Algorithm 1 and (38b) is due to the fact that $\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} = 1$.

From (38c), the following holds

$$\begin{aligned}\mathbb{E}[\|\bar{\mathbf{w}}^{t+1} - \mathbf{w}^*\|_2^2] &= \mathbb{E}[\|\mathbf{w}^t - \eta_t \mathbf{g}^t - \mathbf{w}^* - \eta_t \mathbf{e}^t\|_2^2], \\ &= \mathbb{E}[\|\mathbf{w}^t - \eta_t \mathbf{g}^t - \mathbf{w}^*\|_2^2] - 2\eta_t \mathbb{E}[\langle \mathbf{w}^t - \eta_t \mathbf{g}^t - \mathbf{w}^*, \mathbf{e}^t \rangle] + \eta_t^2 \mathbb{E}[\|\mathbf{e}^t\|_2^2], \\ &\leq \mathbb{E}[\|\mathbf{w}^t - \eta_t \mathbf{g}^t - \mathbf{w}^*\|_2^2] + K\eta_t^2 \mathbb{E}[\|\mathbf{w}^t - \eta_t \mathbf{g}^t - \mathbf{w}^*\|_2^2] + \frac{1}{K} \mathbb{E}[\|\mathbf{e}^t\|_2^2] + \eta_t^2 \mathbb{E}[\|\mathbf{e}^t\|_2^2], \tag{39a} \\ &= (1 + K\eta_t^2) \mathbb{E}[\|\mathbf{w}^t - \eta_t \mathbf{g}^t - \mathbf{w}^*\|_2^2] + \left(\frac{1}{K} + \eta_t^2\right) \mathbb{E}[\|\mathbf{e}^t\|_2^2], \\ &= (1 + K\eta_t^2) \mathbb{E}[\|\mathbf{b}^{L,t} - \mathbf{w}^*\|_2^2] + \left(\frac{1}{K} + \eta_t^2\right) \mathbb{E}[\|\mathbf{e}^t\|_2^2],\end{aligned}\tag{39b}$$

where (39a) is due to the inequality $-2\langle \mathbf{x}, \mathbf{y} \rangle \leq \alpha \|\mathbf{x}\|_2^2 + \frac{1}{\alpha} \|\mathbf{y}\|_2^2$ for $\alpha > 0$ and (39b) follows from the definition of $\{\mathbf{b}^{l,t}\}_{l=0}^L$ in (17).

Next, applying Jensen's inequality to ℓ_2 -norm yields

$$\mathbb{E}[\|\mathbf{e}^t\|_2^2] \leq L \sum_{l=0}^{L-1} \mathbb{E}[\|\mathbf{e}^{l,t}\|_2^2] = L \mathbb{E}[\|\mathbf{e}^{0,t}\|_2^2] + L \sum_{l=1}^{L-1} \mathbb{E}[\|\mathbf{e}^{l,t}\|_2^2]. \quad (40)$$

Regarding the first term on the r.h.s. of (40), we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{e}^{0,t}\|_2^2] &= \mathbb{E} \left[\left\| \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} [\tilde{\nabla} f_{m,u}(\mathbf{w}^{0,m,u,t}) - \nabla f(\mathbf{b}^{0,t})] \right\|_2^2 \right], \\ &= \mathbb{E} \left[\left\| \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} [\tilde{\nabla} f_{m,u}(\mathbf{w}^t) - \nabla f(\mathbf{w}^t)] \right\|_2^2 \right], \\ &= \mathbb{E} \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u}^2 \|\tilde{\nabla} f_{m,u}(\mathbf{w}^t) - \nabla f(\mathbf{w}^t)\|_2^2 \right], \end{aligned} \quad (41a)$$

$$\leq \zeta \mathbb{E}_{\{\mathcal{C}_m\}_{m=1}^M} \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u}^2 \right], \quad (41b)$$

$$\leq \zeta, \quad (41c)$$

where (41a) is due to the unbiasedness in Assumption 3 and the fact that $\{\tilde{\nabla} f_{m,u}(\mathbf{w}^t)\}$ are independent, (41b) follows from the boundedness Assumption 3, and (41c) is due to the fact that $\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u}^2 \leq 1$. Now, the term $\mathbb{E}[\|\mathbf{e}^{l,t}\|_2^2]$ in (40) is rewritten by

$$\begin{aligned} \mathbb{E}[\|\mathbf{e}^{l,t}\|_2^2] &= \mathbb{E} \left[\left\| \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} [\tilde{\nabla} f_{m,u}(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{b}^{l,t})] \right\|_2^2 \right], \\ &= \mathbb{E} \left[\left\| \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} [\tilde{\nabla} f_{m,u}(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{w}^{l,m,u,t}) + \nabla f(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{b}^{l,t})] \right\|_2^2 \right], \\ &= \mathbb{E} \left[\left\| \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} [\tilde{\nabla} f_{m,u}(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{w}^{l,m,u,t})] \right\|_2^2 \right] \\ &\quad + \mathbb{E} \left[\left\| \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} [\nabla f(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{b}^{l,t})] \right\|_2^2 \right], \end{aligned} \quad (42a)$$

where (42a) follows from Assumption 3 that $\mathbb{E}[\tilde{\nabla} f_{m,u}(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{w}^{l,m,u,t})] = \mathbf{0}$. Thus, the

following holds

$$\mathbb{E}[\|\mathbf{e}^{l,t}\|_2^2] \leq \zeta + \mathbb{E} \left[\left(\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u}^2 \right) \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \|\nabla f(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{b}^{l,t})\|_2^2 \right], \quad (43a)$$

$$\leq \zeta + \mathbb{E} \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \|\nabla f(\mathbf{w}^{l,m,u,t}) - \nabla f(\mathbf{b}^{l,t})\|_2^2 \right], \quad (43b)$$

$$\leq \zeta + \mathbb{E}_{\{\mathcal{C}_m\}_{m=1}^M} \left[\mathbb{E} \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \gamma^2 \|\mathbf{w}^{l,m,u,t} - \mathbf{b}^{l,t}\|_2^2 \middle| \{\mathcal{C}_m\}_{m=1}^M \right] \right], \quad (43c)$$

$$= \zeta + \mathbb{E}_{\{\mathcal{C}_m\}_{m=1}^M} \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \mathbb{1}_{\mathcal{C}_m}(d_{m,u}) \gamma^2 \mathbb{E} \left[\|\mathbf{w}^{l,m,u,t} - \mathbf{b}^{l,t}\|_2^2 \right] \right], \quad (43d)$$

$$= \zeta + \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \frac{c_m}{g_m} \gamma^2 \mathbb{E} \left[\|\mathbf{w}^{l,m,u,t} - \mathbf{b}^{l,t}\|_2^2 \right], \quad (43e)$$

$$\leq \zeta + \gamma^2 \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \mathbb{E} \left[\|\mathbf{w}^{l,m,u,t} - \mathbf{b}^{l,t}\|_2^2 \right], \quad (43f)$$

where (43a) follows from $\mathbb{E}[\|\tilde{\nabla} f_{m,u}(\mathbf{x}) - \nabla f(\mathbf{x})\|_2^2] \leq \zeta$ and the result in (10), (43b) is due to the fact that $\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} = 1$, (43c) follows from Assumption 1, (43d) is due to the fact that $\Pr[d_{m,u} \in \mathcal{C}_m] = \frac{1}{g_m}$, (43e) follows from $\frac{c_m}{g_m} \leq 1$, and (43f) is due to the definition of $a^{l,t}$ in (18).

From (19b), plugging (41c) and (43f) into (40) leads to

$$\begin{aligned} \mathbb{E}[\|\mathbf{e}^t\|_2^2] &\leq L\zeta + \sum_{l=1}^{L-1} L \left(\zeta + K\gamma^2 a^{l,t} \right), \\ &= L^2\zeta + LK\gamma^2 \sum_{l=1}^{L-1} a^{l,t}, \\ &\leq L^2\zeta + LK\gamma^2 (L-1) \eta_t^2 L\zeta (1 + L\eta_t^2 \gamma^2)^L, \\ &\leq L^2\zeta + KL^3\gamma^2 \zeta \eta_t^2 e^{L^2\eta_t^2 \gamma^2}, \end{aligned} \quad (44a)$$

$$\leq L^2\zeta + KL^3\gamma^2 \zeta \eta_t^2 e, \quad (44b)$$

where (44a) is due to the fact that $(1+x) \leq e^x$ for $x \geq 0$, and (44b) follows from $\eta_t \leq \frac{1}{L\gamma}$.

Plugging (44b) into (39b) and using (19a) give

$$\mathbb{E}[\|\bar{\mathbf{w}}^{t+1} - \mathbf{w}^*\|_2^2] \leq (1 + K\eta_t^2)(1 - \mu\eta_t)^L \mathbb{E}[\|\mathbf{w}^t - \mathbf{w}^*\|_2^2] + \left(\frac{1}{K} + \eta_t^2 \right) (L^2\zeta + KL^3\gamma^2 \zeta \eta_t^2 e),$$

which completes the proof.

PROOF OF THE UPPER BOUND IN (27a)

The term $\mathbb{E}[\|\check{\mathbf{w}}^{t+1} - \bar{\mathbf{w}}^{t+1}\|_2^2]$ can be bounded as in the following

$$\begin{aligned} \mathbb{E}[\|\check{\mathbf{w}}^{t+1} - \bar{\mathbf{w}}^{t+1}\|_2^2] &= \mathbb{E}\left[\left\|\mathbf{w}^t + \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} \hat{\mathbf{w}}^{m,u,t} - \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} \mathbf{w}^{L,m,u,t}\right\|_2^2\right], \\ &= \mathbb{E}\left[\left\|\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} \mathcal{Q}_{b_m, \epsilon_{1,m,u}}(\hat{\mathbf{v}}^{m,u,t}) - \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} (\mathbf{w}^{L,m,u,t} - \mathbf{w}^t)\right\|_2^2\right], \\ &\leq \mathbb{E}_{\{\mathcal{C}_m\}_{m=1}^M} \left[\mathbb{E} \left[\left(\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u}^2 \right) \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \|\mathcal{Q}_{b_m, \epsilon_{1,m,u}}(\hat{\mathbf{v}}^{m,u,t}) - \mathbf{v}^{m,u,t}\|_2^2 \middle| \{\mathcal{C}_m\}_{m=1}^M \right] \right], \end{aligned} \quad (45a)$$

$$\leq \mathbb{E}_{\{\mathcal{C}_m\}_{m=1}^M} \left[\mathbb{E} \left[\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \|\mathcal{Q}_{b_m, \epsilon_{1,m,u}}(\hat{\mathbf{v}}^{m,u,t}) - \mathbf{v}^{m,u,t}\|_2^2 \middle| \{\mathcal{C}_m\}_{m=1}^M \right] \right], \quad (45b)$$

$$= \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \frac{c_m}{g_m} \mathbb{E} \left[\|\mathcal{Q}_{b_m, \epsilon_{1,m,u}}(\hat{\mathbf{v}}^{m,u,t}) - \hat{\mathbf{v}}^{m,u,t} + \hat{\mathbf{v}}^{m,u,t} - \mathbf{v}^{m,u,t}\|_2^2 \right], \quad (45c)$$

$$\begin{aligned} &\leq 2 \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \frac{c_m}{g_m} \mathbb{E} \left[\|\mathcal{Q}_{b_m, \epsilon_{1,m,u}}(\hat{\mathbf{v}}^{m,u,t}) - \hat{\mathbf{v}}^{m,u,t}\|_2^2 \right] \\ &\quad + 2 \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \frac{c_m}{g_m} \mathbb{E} \left[\left\| \min \left\{ 1, \frac{C}{\|\mathbf{v}^{m,u,t}\|_1} \right\} \mathbf{v}^{m,u,t} - \mathbf{v}^{m,u,t} \right\|_2^2 \right], \quad (45d) \\ &\leq 2 \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \frac{c_m}{g_m} \frac{4dC^2}{(2^{b_m} - 1)^2} + 2\Lambda \sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{G}_m} \frac{c_m}{g_m} \mathbb{E} [\|\mathbf{v}^{m,u,t}\|_2^2], \end{aligned}$$

where (45a) follows from (10) and Step 10 of Algorithm 1, (45b) follows from the fact that

$$\sum_{m=1}^M \sum_{d_{m,u} \in \mathcal{C}_m} \omega_{m,u} = 1;$$

(45c) follows from

$$\Pr[d_{m,u} \in \mathcal{C}_m] = \frac{1}{g_m}, \quad \forall d_{m,u} \in \mathcal{G}_m;$$

(45d) follows from the arithmetic-geometric mean (AGM) inequality and Step 11 of Algorithm 1;

and the last inequality follows from Assumption 4 and from the fact that, for $\mathbf{x} \in \mathbb{R}^{d \times 1}$ with

$$\|\mathbf{x}\|_1 \leq C$$

$$\mathbb{E}[\|\mathcal{Q}_{b, \epsilon_1}(\mathbf{x}) - \mathbf{x}\|_2^2] = \sum_{i=1}^d \mathbb{E}[|\mathcal{Q}_{b, \epsilon_1}(x_i) - x_i|^2] \leq \frac{4dC^2}{(2^b - 1)^2}.$$

The last bound is due to the distortion in (8) and the fact that

$$\max \{ (q_i - a)^2, (q_{i+1} - a)^2 \} \leq \frac{(\bar{a} - \underline{a})^2}{(2^b - 1)^2}.$$